

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		add soap solution to each sample and shake (1) the sample that produces a lather is soft water (1) boil the remaining samples (1) the sample that now gives a lather is temporary hard water and the one that still does not give a lather is permanent hard water (1)	2		2	4		4
		(ii)		Z is temporary hard water / has significant temporary hardness with a small amount of permanent hardness (1) it has a high concentration of hydrogencarbonate ions / it has a high concentration of hydrogencarbonate ions and a small concentration of sulfate ions (1) ignore reference to high magnesium ion concentration			2	2		